MTH 200 Chapter 3: Derivatives

3.3.2 Derivatives Worksheets

****Remember that simplifying before attempting to find the derivative is worthwhile****

1. Using Leibniz notation $\left(i.e., \frac{dy}{dx}, \frac{dP}{dt}, etc. \right)$, find the derivatives for each of the following functions, using the Product Rule when possible.

a) $y = e^x \left(\frac{1}{2}x^2 + 2x \right)$

b) $h(t) = (t-2)(2t+3)$

c) $g(u) = \left(\frac{1}{u^2} - \frac{3}{u^4} \right)(u + 5u^3)$

d) $y = x^{-1}e^x \left(x^4 - 2x + \pi \right)$

e) $w(r) = (1 + r + r^2)(2 - r^4)$

f) $A(s)=\sqrt[3]{s}\left(s^2+s+s^{-1}\right)$

2. Find the equation of the tangent line to the curve at the given value of x.

a) $y=(x+1)e^x$, $x=0$

b) $y=\left(x^2-3x+2\right)\left(x^3+x+1\right)$, $x=1$

c) $y=e^x\sqrt{x}\left(x^2-x+2\right)$, $x=1$

3. Consider the function $y=\left(x^2-3\right)e^x$.

a) Determine all value of x at which the function has horizontal tangent lines.

b) Write the equations of all horizontal tangent lines found in part (a).

4. Determine the value of $h'(2)$, given that $f(2)=-3,\ g(2)=4,\ f'(2)=-2,\ \&\ g'(2)=7$.

a) $h(x)=5f(x)-4g(x)$

b) $h(x)=f(x)g(x)$

5. If $f(x)=\sqrt{x}g(x)e^x$ where $g(4)=8$ and $g'(4)=7$, find $f'(4)$.